COHOMOLOGY OF A FILTERED COLIMIT OF PULLED BACK SHEAVES AS THE COLIMIT OF THE COHOMOLOGIES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement begins 'In the situation discussed above.', referring to the setup of the preceding paragraph of the chapter; the counted span reproduces the statement as written and the setup itself is a navigation pointer carried as a stub.

Editorial note. The counted spans carry 8 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-cohomology/retained-preamble.tex lines 1-81), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-colimit is defined only in cohomology.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Filtered colimits are exact. Algebra, Lemma algebra-lemma-directed-colimit-exact, invoked at proof lines 3427-3428. Supplies that the colimit of the acyclic Cech complexes is acyclic in positive degrees.
- Products of injective sheaves are injective. Homology, Lemma homology-lemma-product-injectives, invoked at proof lines 3400-3401. Supplies that the sheaves G_i built as products of pushforwards are injective.
- Pushforward along a flat morphism of ringed spaces preserves injectives. Same chapter, Lemma lemma-pushforward-injective-flat, invoked at proof lines 3399-3400. Supplies that the individual pushforwards f_{b,*}I_k are injective on X_i.
- Vanishing criterion by Cech cohomology on a basis. Same chapter, Lemma lemma-cech-vanish-basis, invoked at proof lines 3413-3415. Supplies that vanishing of Cech cohomology for coverings of quasi-compact opens by finitely many quasi-compact opens implies vanishing of cohomology.
- Open coverings of a quasi-compact open descend to some X_i. Topology, Lemma topology-lemma-descend-opens, invoked at proof lines 3416-3417. Supplies that such a covering is the inverse image of a covering on some X_i.

Source reference. Original lemma 00DB: sources/stacks-chow/tags.txt, source lines 491–491. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 0137: sources/stacks-chow/tags.txt, source lines 1425–1425. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01EW: sources/stacks-cohomology/pinned-original-source.tex, source lines 1696–1696. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01EP: sources/stacks-cohomology/pinned-original-source.tex, source lines 1408–1408. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02N5: sources/stacks-cohomology/pinned-original-source.tex, source lines 1821–1821. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01FF: sources/stacks-cohomology/pinned-original-source.tex, source lines 3194–3194. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0A33: sources/stacks-chow/tags.txt, source lines 12726–12726. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0A30: sources/stacks-chow/tags.txt, source lines 12723–12723. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *In the situation discussed above. Let $i \in \mathrm{Ob}(\mathcal{I})$ and let $U_i \subset X_i$ be quasi-compact open. Then*

$$\mathrm{colim}_{a:j\to i} H^p(f_a^{-1}(U_i), \mathcal{F}_j) = H^p(p_i^{-1}(U_i), \mathcal{F})$$

for all $p \geq 0$. In particular we have $H^p(X, \mathcal{F}) = \mathrm{colim}\, H^p(X_i, \mathcal{F}_i)$.

Proof. The case $p = 0$ is Sheaves, Lemma 0A33.

In this paragraph we show that we can find a map of systems $(\gamma_i) : (\mathcal{F}_i, \varphi_a) \to (\mathcal{G}_i, \psi_a)$ with $\mathcal{G}_i$ an injective abelian sheaf and γ_i injective. For each i we pick an injection $\mathcal{F}_i \to \mathcal{I}_i$ where $\mathcal{I}_i$ is an injective abelian sheaf on X_i. Then we can consider the family of maps

$$\gamma_i : \mathcal{F}_i \longrightarrow \prod\nolimits_{b:k\to i} f_{b,*}\mathcal{I}_k = \mathcal{G}_i$$

where the component maps are the maps adjoint to the maps $f_b^{-1}\mathcal{F}_i \to \mathcal{F}_k \to \mathcal{I}_k$. For $a : j \to i$ in $\mathcal{I}$ there is a canonical map

$$\psi_a : f_a^{-1}\mathcal{G}_i \to \mathcal{G}_j$$

whose components are the canonical maps $f_b^{-1} f_{a\circ b,*}\mathcal{I}_k \to f_{b,*}\mathcal{I}_k$ for $b : k \to j$. Thus we find an injection $\{\gamma_i\} : \{\mathcal{F}_i, \varphi_a) \to (\mathcal{G}_i, \psi_a)$ of systems of abelian sheaves. Note that $\mathcal{G}_i$ is an injective sheaf of abelian groups on X_i, see Lemma 02N5 and Homology, Lemma 0137. This finishes the construction.

Arguing exactly as in the proof of Lemma 01FF we see that it suffices to prove that $H^p(X, \operatorname{colim} f_i^{-1}\mathcal{G}_i) = 0$ for $p > 0$.

Set $\mathcal{G} = \operatorname{colim} f_i^{-1}\mathcal{G}_i$. To show vanishing of cohomology of $\mathcal{G}$ on every quasi-compact open of X, it suffices to show that the Čech cohomology of $\mathcal{G}$ for any open covering $\mathcal{U}$ of a quasi-compact open of X by finitely many quasi-compact opens is zero, see Lemma 01EW. Such a covering is the inverse by p_i of such a covering $\mathcal{U}_i$ on the space X_i for some i by Topology, Lemma 0A30. We have

$$\check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{G}) = \operatorname{colim}_{a:j\to i} \check{\mathcal{C}}^\bullet(f_a^{-1}(\mathcal{U}_i), \mathcal{G}_j)$$

by the case $p = 0$. The right hand side is a filtered colimit of complexes each of which is acyclic in positive degrees by Lemma 01EP. Thus we conclude by Algebra, Lemma 00DB. □